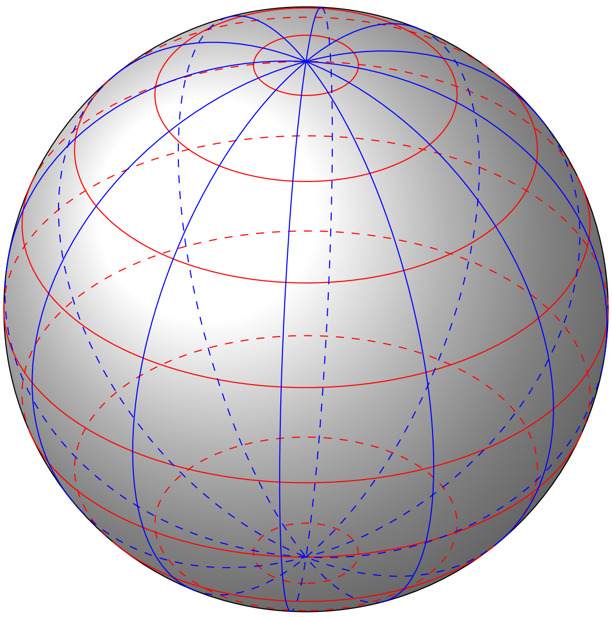
MATH30106

แคลคูลัสและการประยุกต์
Calculus and Its Applications

Handout



1

Sequences and Series

1.1 Sequences and The Partial Sums of a Sequence

Roughly speaking, a sequence is an infinite list of numbers. The numbers in the sequence are often written as $a_1, a_2, a_3, \ldots$. The dots mean that the list continues forever.

1.1.1 Sequences

Any ordered list of numbers can be viewed as a function whose input values are $1, 2, 3, \ldots$ and whose output values are the numbers in the list. So we define a sequence as follows.

Definition 1.1.1

A **sequence** (ลำดับ) is a function a whose domain is the set of positive integers. The terms of the sequence are the function values

$$a(1), a(2), a(3), \ldots, a(n), \ldots$$

We usually write a_n instead of the function notation $a(n)$. So the terms of the sequence are written as

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

The number a_1 is called the **first term**, a_2 is called the **second term**, and in general, a_n is called the n**th term** (พจน์ที่ n).

On occasion it is convenient to begin subscripting a sequence with 0 instead of 1 so that the terms of the sequence become $a_0, a_1, a_2, a_3, \ldots, a_n, \ldots$.

Example 1.1.2

Find the first five terms and the 100th term of the sequence defined by each formula.

(a) $a_n = 2n - 1$

(b) $c_n = n^2 - 1$

(c) $t_n = \dfrac{n}{n+1}$

(d) $r_n = \dfrac{(-1)^n}{2^n}$

Solution. To find the first five terms, we substitute $n = 1, 2, 3, 4$ and 5 in the formula for the nth term. To find the 100th term, we substitute $n = 100$. This gives the following.

nth term	First five terms	100th term
(a) $2n - 1$	$1, 3, 5, 7, 9$	199
(b) $n^2 - 1$	$0, 3, 8, 15, 24$	9999
(c) $\dfrac{n}{n+1}$	$\dfrac{1}{2}, \dfrac{2}{3}, \dfrac{3}{4}, \dfrac{4}{5}, \dfrac{5}{6}$	$\dfrac{100}{101}$
(d) $\dfrac{(-1)^n}{2^n}$	$-\dfrac{1}{2}, \dfrac{1}{4}, -\dfrac{1}{8}, \dfrac{1}{16}, -\dfrac{1}{32}$	$\dfrac{1}{2^{100}}$

□

In Example 1.1.2(d) the presence of $(-1)^n$ in the sequence has the effect of making successive terms alternately negative and positive.

It is often useful to picture a sequence by sketching its graph. Since a sequence is a function whose domain is the natural numbers, we can draw its graph in the Cartesian plane. For instance, the graph of the sequence

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{n}, \dots$$

is shown in Figure 1.1. Compare the graph of the sequence shown in Figure 1.1 to the graph of

$$1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots, \frac{(-1)^{n+1}}{n}, \dots$$

shown in Figure 1.2.

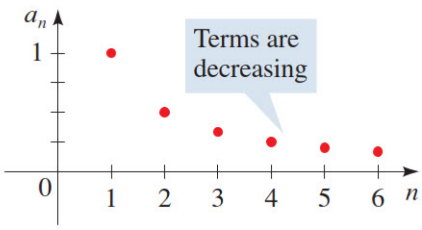


Figure 1.1

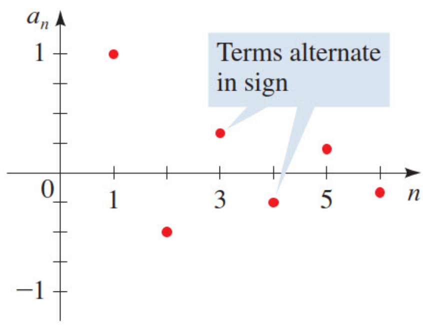


Figure 1.2

The graph of every sequence consists of isolated points that are not connected.

Example 1.1.3

Write an expression for the apparent nth term (a_n) of each sequence.

1. $\frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \frac{7}{8}, \ldots$
2. $-2, 4, -8, 16, -32, \ldots$

Some sequences are defined **recursively**. To define a sequence recursively, you need to be given one or more of the first few terms. All other terms of the sequence are then defined using previous terms. A well-known example is the Fibonacci sequence.

Example 1.1.4

The Fibonacci sequence is defined recursively, as follows.

$$F_0 = 1,\ F_1 = 1,\ F_k = F_{k-2} + F_{k-1},\ \text{ where } k \geq 2$$

Write the first eleven terms of this sequence.

The sequence in Example 1.1.4 is called the **Fibonacci sequence**, named after the 13th century Italian mathematician who used it to solve a problem about the breeding of rabbits. The sequence also occurs in numerous other applications in nature. In fact, so many phenomena behave like the Fibonacci sequence that one mathematical journal, the *Fibonacci Quarterly*, is devoted entirely to its properties.

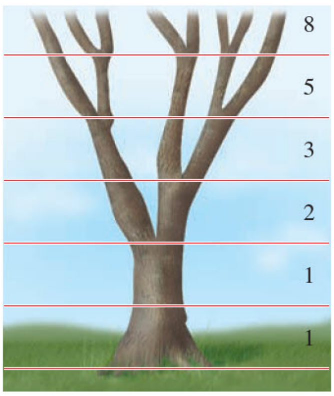
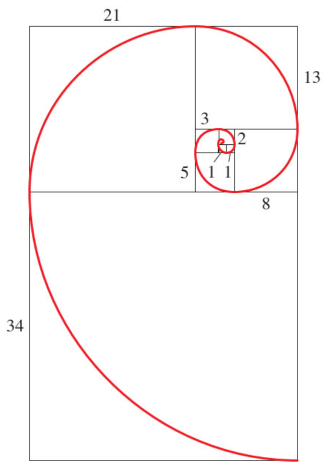
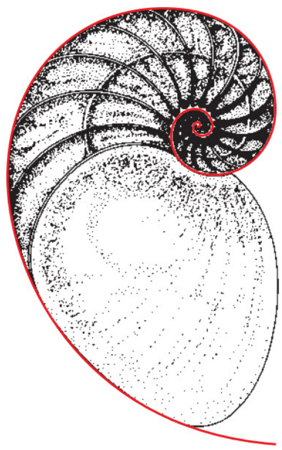


Figure 1.3: The Fibonacci sequence in the branching of a tree



Fibonacci spiral



Nautilus shell

1.1.2 Summation Notation

There is a convenient notation for the sum of the terms of a finite sequence. It is called **summation notation** or **sigma notation** because it involves the use of the uppercase Greek letter sigma, written as $\sum$.

Definition 1.1.5

The sum of the first n terms of a sequence is represented by

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + \ldots + a_n$$

where k is called the **index of summation**, n is the **upper limit of summation**, and 1 is the **lower limit of summation**.

Example 1.1.6

$$\sum_{k=0}^{8} \frac{1}{k!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} \approx 2.71828$$

For this summation, note that the sum is very close to the irrational number $e \approx 2.718281828$. It can be shown that as more terms of the sequence whose nth term is $1/n!$ are added, the sum becomes closer and closer to e.

Example 1.1.7

$$\sqrt{3} + \sqrt{4} + \sqrt{5} + \ldots + \sqrt{77} = \sum_{k=3}^{77} \sqrt{k}$$

However, there is no unique way of writing a sum in sigma notation. We could also write this sum as

$$\sqrt{3} + \sqrt{4} + \sqrt{5} + \ldots + \sqrt{77} = \sum_{k=0}^{74} \sqrt{k+3} = \sum_{k=1}^{75} \sqrt{k+2}.$$

Theorem 1.1.8: Properties of Sums

1. $\displaystyle\sum_{k=1}^{n} c = cn,$ $\quad c$ is a constant.
2. $\displaystyle\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k,$ $\quad c$ is a constant.
3. $\displaystyle\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$
4. $\displaystyle\sum_{k=1}^{n} (a_k - b_k) = \sum_{k=1}^{n} a_k - \sum_{k=1}^{n} b_k$

Example 1.1.9

The arithmetic mean $\bar{x}$ of a set of n measurements $x_1, x_2, ..., x_n$ is defined as $\bar{x} = \dfrac{1}{n}\sum_{i=1}^{n} x_i$.

Prove that

(a) $\displaystyle\sum_{i=1}^{n}(x_i - \bar{x}) = 0$

(b) $\displaystyle\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2$

Theorem 1.1.10

(a) $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

(b) $\displaystyle\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$

(c) $\displaystyle\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2 = \left(\sum_{i=1}^{n} i\right)^2$

Proof.

(a) Let $S_n = 1 + 2 + 3 + ... + (n-1) + n$. As $S_n = n + (n-1) + ... + 2 + 1$, we get

$$2S_n = \underbrace{(n+1) + (n+1) + ... + (n+1)}_{n \text{ terms}}$$

Hence, $S_n = \dfrac{n(n+1)}{2}$.

(b) Observe that $i^3 - (i-1)^3 = 3i^2 - 3i + 1$ and

$$\sum_{i=1}^{n} \left[i^3 - (i-1)^3\right] = \left(1^3 - 0^3\right) + \left(2^3 - 1^3\right) + \left(3^3 - 2^3\right) + \dots + \left(n^3 - (n-1)^3\right) = n^3.$$

It follows from above equations that

$$\begin{aligned} n^3 &= \sum_{i=1}^{n} \left(3i^2 - 3i + 1\right) \\ &= 3\sum_{i=1}^{n} i^2 - 3\sum_{i=1}^{n} i + \sum_{i=1}^{n} 1 \\ &= 3\sum_{i=1}^{n} i^2 - 3\left[\frac{n(n+1)}{2}\right] + n. \end{aligned}$$

Therefore,

$$\sum_{i=1}^{n} i^2 = \frac{1}{3}\left(n^3 + \frac{3}{2}n(n+1) - n\right) = \frac{n(n+1)(2n+1)}{6}.$$

(c) Using the fact that $i^4 - (i-1)^4 = 4i^3 - 6i^2 + 4i - 1$ and

$$\sum_{i=1}^{n} \left[i^4 - (i-1)^4\right] = \left(1^4 - 0^4\right) + \left(2^4 - 1^4\right) + \left(3^4 - 2^4\right) + \dots + \left(n^4 - (n-1)^4\right) = n^4,$$

the desired identity can be proved in the same manner as (b).

$\square$

Example 1.1.11

Find the sum: $1 \cdot 2 + 3 \cdot 4 + 5 \cdot 6 + \dots + (2n-1) \cdot (2n)$.

1.1.3 The Partial Sums of a Sequence

In calculus we are often interested in adding the terms of a sequence. This leads to the following definition.

Definition 1.1.12

Consider the sequence $a_1, a_2, a_3, ..., a_n, ...$, the **partial sum** are

$$\begin{aligned} S_1 &= a_1 \\ S_2 &= a_1 + a_2 = \sum_{k=1}^{2} a_k \\ S_3 &= a_1 + a_2 + a_3 = \sum_{k=1}^{3} a_k \\ &\vdots \\ S_n &= a_1 + a_2 + a_3 + ... + a_n = \sum_{k=1}^{n} a_k \\ &\vdots \end{aligned}$$

S_1 is called the **first partial sum**, S_2 is the **second partial sum**, and so on. S_n is called the n**th partial sum** (ผลบวกย่อยอันดับที่ n หรือ ผลบวก n พจน์แรก). The sequence $S_1, S_2, S_3, ..., S_n, ...$ is called the **sequence of partial sums** (ลำดับของผลบวกย่อย).

Example 1.1.13

Find the first four partial sums and the nth partial sum of the sequence given by $a_n = 1/2^n$.

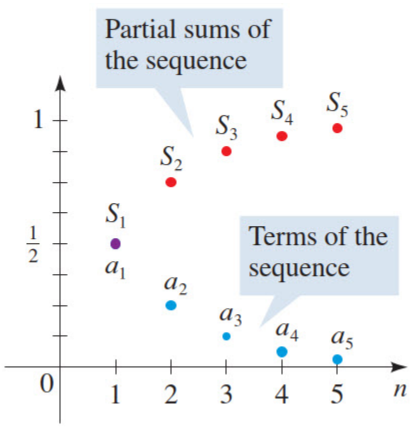


Figure 1.4: Graph of the sequence a_n and the sequence of partial sum S_n

1.1.4 Arithmetic Sequences and Partial Sums

Arithmetic Sequences

A sequence whose consecutive terms have a common difference is called an arithmetic sequence.

Definition 1.1.14

An **arithmetic sequence** (ลำดับเลขคณิต) is a sequence of the form

$$a,\ a+d,\ a+2d,\ a+3d,\ a+4d, \ldots \quad (d \text{ is a fixed constant}).$$

The number a is the **first term**, and d is the **common difference** (ผลต่างร่วม) of the sequence. The n**th term** of an arithmetic sequence is given by

$$a_n = a + (n-1)d$$

The number d is called the common difference because any two consecutive terms of an arithmetic sequence differ by d.

Example 1.1.15

1. If $a = 2$ and $d = 3$, then we have the arithmetic sequence

$$2,\ 2+3,\ 2+6,\ 2+9, \ldots$$

or $2,\ 5,\ 8,\ 11, \ldots$. Any two consecutive terms of this sequence differ by $d = 3$. The nth term is $a_n = 2 + 3(n-1)$.

2. Consider the arithmetic sequence

$$9,\ 4,\ -1,\ -6,\ -11, \ldots$$

Here the common difference is $d = -5$. The terms of an arithmetic sequence decrease if the common difference is negative. The nth term is $a_n = \boxed{}$

3. The graph of the arithmetic sequence $a_n = 1 + 2(n-1)$ is shown in Figure 1.5. Notice that the points in the graph lie on the straight line $y = 2x - 1$, which has slope $d = 2$.

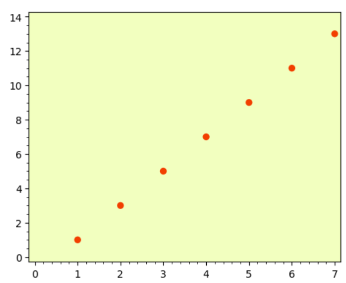


Figure 1.5

Example 1.1.16

The 11th term of an arithmetic sequence is 52, and the 19th term is 92. Find the 1000th term.

Remark 1.1.17

If $a_1, a_2, a_3, ..., a_n, ...$ is an arithmetic sequence with common difference d, then

$$a_n = a_k + (n-k)d$$

for any positive integer k.

Partial Sums of Arithmetic Sequences

We want to find the sum of the first n terms of the arithmetic sequence

$$a,\ a+d,\ a+2d,\ a+3d, ..., a+(n-1)d, ...$$

We write (using Gauss's method)

$$S_n = a + (a+d) + (a+2d) + (a+3d) + ... + (a+(n-1)d) \tag{1.1}$$

and

$$S_n = (a+(n-1)d) + (a+(n-2)d) + (a+(n-3)d) + ... + (a+d) + a. \tag{1.2}$$

Adding (1.1) and (1.2 gives

$$2S_n = (2a+(n-1)d) + (2a+(n-1)d) + ... + (2a+(n-1)d). \tag{1.3}$$

There are n identical terms on the right side of this equation, so

$$S_n = \frac{n}{2}(2a+(n-1)d).$$

Notice that $a_n = a + (n-1)d$ is the nth term of this sequence. So we can write

$$S_n = \frac{n}{2}(2a + (n-1)d) = n\left(\frac{a + a_n}{2}\right).$$

This last formula says that the sum of the first n terms of an arithmetic sequence is the average of the first and nth terms multiplied by n, the number of terms in the sum. We now summarize this result.

Theorem 1.1.18: Partial Sums of an Arithmetic Sequence

For the arithmetic sequence given by $a_n = a + (n-1)d$, the n**th partial sum**

$$S_n = a + (a+d) + (a+2d) + (a+3d) + \ldots + (a + (n-1)d)$$

is given by either of the following formulas.

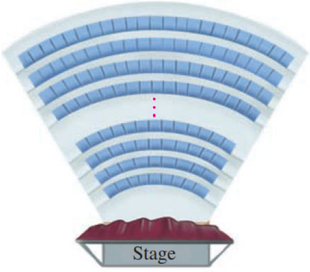
1. $S_n = \dfrac{n}{2}(2a + (n-1)d)$
2. $S_n = n\left(\dfrac{a + a_n}{2}\right)$

Example 1.1.19

Find the sum of the first n odd numbers.

Example 1.1.20

Find the sum: $3 + 7 + 11 + 15 + \ldots + 159$.



Example 1.1.21

An amphitheater has 50 rows of seats with 30 seats in the first row, 32 in the second, 34 in the third, and so on. Find the total number of seats.

Example 1.1.22

Let $a_1, a_2, ..., a_{100}$ be an arithmetic sequence such that $5a_{51} = a_{53} + 16$. Find $a_1 + a_2 + ... + a_{100}$.

Example 1.1.23

If the nth partial sum of an arithmetic sequence is $3n^2 + 2n$, find the nth term of this sequence.

1.1.5 Geometric Sequences and and Partial Sums

Geometric Sequences

In this section we study geometric sequences. This type of sequence occurs frequently in applications to finance, population growth, and other fields.

Definition 1.1.24

A **geometric sequence** (ลำดับเรขาคณิต) is a sequence of the form

$$a, ar, ar^2, ar^3, ar^4, \ldots \qquad (r \text{ is a fixed nonzero constant}).$$

The number a is the **first term**, and r is the **common ratio** (อัตราส่วนร่วม) of the sequence. The n**th term** of a geometric sequence is given by

$$a_n = ar^{n-1}.$$

The number r is called the common ratio because the ratio of any two consecutive terms of the sequence is r.

Example 1.1.25

1. If $a = 3$ and $r = 2$, then we have the geometric sequence
$$3,\ 3 \cdot 2,\ 3 \cdot 2^2,\ 3 \cdot 2^3,\ 3 \cdot 2^4, \ldots$$
or $3,\ 6,\ 12,\ 24,\ 48, \ldots$. Notice that the ratio of any two consecutive terms is $r = 2$. The nth term is $a_n = 3 \cdot 2^{n-1}$.
2. The sequence
$$2,\ -10,\ 50,\ -250,\ 1250, \ldots$$
is a geometric sequence with $a = 2$ and $r = -5$. When r is negative, the terms of the sequence alternate in sign. The nth term is $a_n = 2(-5)^{n-1}$.
3. The sequence
$$1,\ \frac{1}{3},\ \frac{1}{9},\ \frac{1}{27},\ \frac{1}{81}, \ldots$$
is a geometric sequence with $a = 1$ and $r = \frac{1}{3}$ (see Figure 1.6). The nth term is $a_n = 1\left(\frac{1}{3}\right)^{n-1}$.
4. The graph of the geometric sequence $a_n = \frac{1}{5} \cdot 2^{n-1}$ is shown in Figure 1.7. Notice that the points in the graph lie on the graph of the exponential function $\frac{1}{5} \cdot 2^{x-1}$.

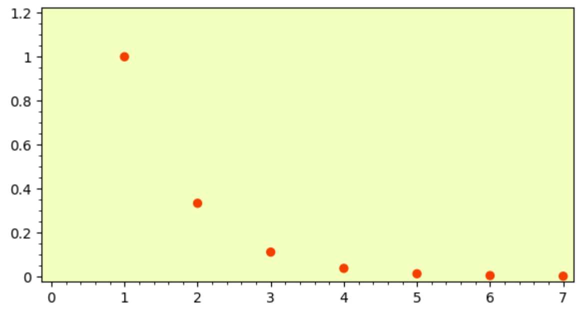


Figure 1.6: A graph of $a_n = \left(\frac{1}{3}\right)^{n-1}$

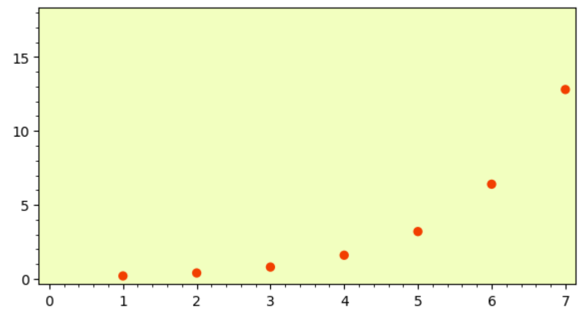


Figure 1.7: A graph of $a_n = \frac{1}{5} \cdot 2^{n-1}$

Remark 1.1.26

If $0 < r < 1$, then the terms of the geometric sequence ar^{n-1} decrease, but if $r > 1$, then the terms increase. (What happens if $r = 1$?)

Example 1.1.27

Suppose a ball has elasticity such that when it is dropped, it bounces up one-third of the distance it has fallen. If this ball is dropped from a height of 2 m, then it bounces up to a height of $\boxed{}$ m. On its second bounce, it returns to a height of $\boxed{}$ m, and so on (see Figure 1.8). Thus, the height h_n that the ball reaches on its nth bounce is given by the geometric sequence

$$h_n = \boxed{}$$

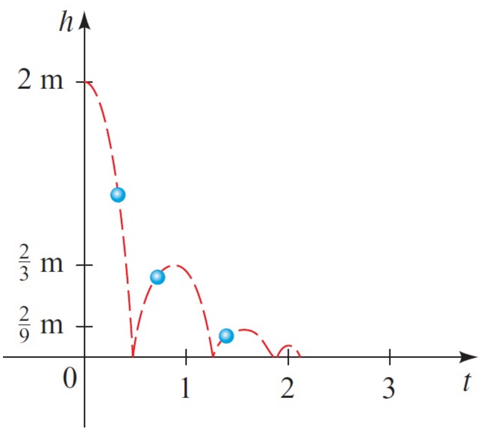


Figure 1.8

Example 1.1.28

The third term of a geometric sequence is $\frac{63}{4}$, and the sixth term is $\frac{1701}{32}$. Find the fifth term.

Remark 1.1.29

If $a_1, a_2, a_3, \ldots, a_n, \ldots$ is an geometric sequence with common ratio r, then

$$a_n = a_k r^{n-k}$$

for any positive integer k.

Partial Sums of Geometric Sequences

For the geometric sequence $a, ar, ar^2, ar^3, ar^4, \ldots, ar^{n-1}, \ldots$, the nth partial sum is

$$S_n = \sum_{k=1}^{n} ar^{k-1} = a + ar + ar^2 + ar^3 + \ldots + ar^{n-1}$$

To find a formula for S_n, we multiply S_n by r and subtract from S_n.

$$S_n = a + ar + ar^2 + ar^3 + \ldots + ar^{n-1} \tag{1.4}$$

$$rS_n = ar + ar^2 + ar^3 + \ldots + ar^{n-1} + ar^n \tag{1.5}$$

It follows that $S_n(1-r) = a\,(1-r^n)$, and hence $S_n = \dfrac{a\,(1-r^n)}{1-r}$.

Theorem 1.1.30: Partial Sums of a Geometric Sequence

For the geometric sequence $a_n = ar^{n-1}$, the n**th partial sum** is given by

$$S_n = \frac{a\,(1-r^n)}{1-r}.$$

Example 1.1.31

Find the sum of the first five terms of the geometric sequence

$$1,\ 0.7,\ 0.49,\ 0.343, \ldots$$

Example 1.1.32

Find the sum:

(a) $9 + 99 + 999 + \ldots + \underbrace{999\ldots9}_{n \text{ terms}}$

(b) $1 + (1+2) + (1+2+2^2) + \ldots + \left(1+2+2^2+\ldots+2^n\right)$

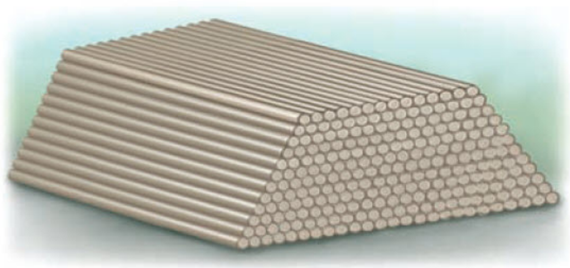
1.1.6 Exercises

1. Find the product of the numbers

$$10^{1/10},\ 10^{2/10},\ 10^{3/10},\ldots,\ 10^{19/10}$$

2. An arithmetic sequence has first term $a_1 = 1$ and fourth term $a_4 = 16$. How many terms of this sequence must be added to get 2356?

3. Telephone poles are being stored in a pile with 25 poles in the first layer, 24 in the second, and so on. If there are 12 layers, how many telephone poles does the pile contain?



4. An architect designs a theater with 15 seats in the first row, 18 in the second, 21 in the third, and so on. If the theater is to have a seating capacity of 870, how many rows must the architect use in his design?

5. The second and fifth terms of a geometric sequence are 10 and 1250, respectively. Is $31,250$ a term of this sequence? If so, which term is it?

6. Find the sum of the first ten terms of the sequence

$$a + b,\ a^2 + 2b,\ a^3 + 3b,\ a^4 + 4b,\ldots$$

7. A ball is dropped from a height of 9 ft. The elasticity of the ball is such that it always bounces up one-third the distance it has fallen. Find a formula for the total distance the ball has traveled at the instant it hits the ground the nth time.

8. The following is a well-known children's rhyme:

   As I was going to St. Ives,
   I met a man with seven wives;
   Every wife had seven sacks;
   Every sack had seven cats;
   Every cat had seven kits;
   Kits, cats, sacks, and wives,
   How many were going to St. Ives?

   Assuming that the entire group is actually going to St. Ives, show that the answer to the question in the rhyme is a partial sum of a geometric sequence, and find the sum.

9. กำหนดให้ $\{a_n\}$ เป็นลำดับโดยที่ $a_1 = 2$ และ $a_n = \sqrt{a_{n-1}}$ สำหรับ $n = 2, 3, 4, ...$ จงหาพจน์ทั่วไปของลำดับ $\{a_n\}$
$\left(\textit{Hint}: \text{ พิจารณาลำดับ } b_n = \ln(a_n)\right)$

10. กำหนดให้ $\{a_n\}$ เป็นลำดับโดยที่ $a_1 = 1, a_2 = 3, a_3 = 7, a_4 = 13, a_5 = 21, ...$ จงหาพจน์ทั่วไปของลำดับ $\{a_n\}$
$\left(\textit{Hint}: \text{ พิจารณาลำดับ } b_n = a_{n+1} - a_n\right)$

11. จงหาพจน์ 5 พจน์ของลำดับเลขคณิตที่อยู่ระหว่าง 9 และ 34

12. กำหนดให้ $A = \{20, 21, 22, ..., 1,000\}$

    12.1. ตัวเลขในเซต A ที่หารด้วย 3 ลงตัวมีกี่จำนวน

    12.2. ตัวเลขในเซต A ที่หารด้วย 5 ลงตัวมีกี่จำนวน

    12.3. ตัวเลขในเซต A ที่หารด้วย 3 และ 5 ลงตัวมีกี่จำนวน

    12.4. ตัวเลขในเซต A ที่หารด้วย 7 ไม่ลงตัวมีกี่จำนวน

13. พจน์แรกที่เป็นจำนวนเต็มลบของลำดับ $200, 182, 164, 146, ...$ มีค่าต่างจากพจน์ที่ 15 เท่ากับเท่าใด

14. จงหาค่า k ที่ทำให้รากของสมการ $x^3 + 3x^2 - 6x + k = 0$ เป็นลำดับเลขคณิต

15. A และ B อยู่ห่างกัน 255 กิโลเมตร และตั้งใจจะเดินทางมาพบกันโดยออกเดินทาง พร้อมกัน A เดินทางวันแรก 1 กิโลเมตร วันที่สอง 3 กิโลเมตร วันที่สาม 5 กิโลเมตร เช่นนี้ไปเรื่อย ๆ ส่วน B เดินทางวันแรก 20 กิโลเมตร วันที่สอง 19 กิโลเมตร วันที่สาม 18 กิโลเมตร เช่นนี้ไปเรื่อย ๆ จงหาว่าในเวลากี่วันเขาทั้งสองจะพบกัน

16. กำหนดให้ a, b, c เป็นสามพจน์เรียงกันในลำดับเรขาคณิต และมีผลคูณเป็น 27
ถ้า $a, b + 3, c + 2$ เป็นสามพจน์เรียงติดกันในลำดับเลขคณิต แล้ว $a + b + c$ มีค่าเท่าใด

17. ถ้า n เป็นจำนวนเต็มบวกที่ทำให้ผลบวก n พจน์แรกของอนุกรมเลขคณิต $7 + 15 + 23 + ...$ มีค่าเท่ากับ 217 แล้ว
$$\frac{2^n + 2^{n+1} + ... + 2^{2n}}{2^8}$$
เท่ากับเท่าใด

18. กำหนดให้ $\{a_n\}$ เป็นลำดับเลขคณิตซึ่งมีผลต่างร่วมไม่เท่ากับศูนย์
ถ้า $\displaystyle\sum_{n=100}^{200} a_n = c \sum_{n=100}^{200} (-1)^n a_n \neq 0$ แล้ว c มีค่าเท่าใด

19. จงหาจำนวนลำดับเลขคณิต a_1, a_2, a_3 ที่สอดคล้องสมบัติทั้ง 2 ข้อต่อไปนี้

    (1) $a_1 < a_2 < a_3$

    (2) $a_1, a_2, a_3 \in \{1, 2, 3, ..., 19\}$

20. กำหนดให้ $\{a_n\}$ เป็นลำดับของจำนวนจริง โดยที่ $a_1 = a_2 = a_3 = 1, a_4 = 2$ สำหรับจำนวนเต็มบวก n ใด ๆ $a_n a_{n+1} a_{n+2} a_{n+3} \neq 1$ และ

$$a_n a_{n+1} a_{n+2} a_{n+3} a_{n+4} = a_n + a_{n+1} + a_{n+2} + a_{n+3} + a_{n+4}$$

จงหาค่าของ $\sum_{k=1}^{100} a_k$

21. ให้ $a_1, a_2, ..., a_n$ เป็นลำดับเลขคณิตซึ่ง $a_4 + a_7 + a_{10} = 17$ และ $\sum_{i=4}^{14} a_i = 77$
ถ้า $a_n = 13$ และ $\dfrac{1+3+5+...+(2n-1)}{a_1 + a_2 + ... + a_n} = \dfrac{r}{s}$ เมื่อ r และ s เป็นจำนวน เต็มบวกที่ $(r, s) = 1$ แล้ว ค่าของ $r + s$ เท่ากับเท่าใด

22. กำหนดให้ $a_1, a_2, ..., a_n$ เป็นลำดับของจำนวนจริงซึ่ง $a_1 = 17$ และ $a_{n+1} = 2a_n - 7$ ทุก $n = 1, 2, ...$ สมมติว่ามีจำนวนจริง r, s และ t ที่ทำให้ $a_n = r(s^n) + t$ สำหรับทุก $n = 1, 2, ...$ จงหาค่าของ $r + s + t$

23. จงหาผลบวก n พจน์แรกของอนุกรม $4 + 44 + 444 + ... + \underbrace{444...4}_{n \text{ ตัว}}$

24. จงหาค่าของ $\sum_{k=1}^{999} \sqrt{1 + \dfrac{1}{k^2} + \dfrac{1}{(k+1)^2}}$
$\left(\textit{\textbf{Hint}} : 1 + \dfrac{1}{k^2} + \dfrac{1}{(k+1)^2} = 1 + \dfrac{2}{k(k+1)} + \dfrac{1}{k^2} - \dfrac{2}{k(k+1)} + \dfrac{1}{(k+1)^2} \right)$

25. กำหนดให้ $\{a_n\}$ เป็นลำดับของจำนวนจริงบวก โดยที่ $a_i \neq 1$ ทุกจำนวนเต็มบวก i ถ้า $a_0 = 2, a_1 = 4$ และ

$$(a_n)^{\log a_n} = (a_{n+1})^{\log a_{n-1}} \quad \text{สำหรับทุก } n = 1, 2, 3, ...$$

จงหาพจน์ที่ n ของลำดับนี้

26. ถ้า $\{a_n\}$ เป็นลำดับซึ่ง $a_n > 0$ และ $\dfrac{a_{n+1}^2}{a_{n+1} + 2a_n} = a_n$ สำหรับทุกจำนวนนับ n แล้ว $\dfrac{1}{a_1} \sum_{n=1}^{10} a_n$ มีค่าเท่าใด

27. ถ้า $a_1, a_2, ...$ และ $b_1, b_2, ...$ เป็นลำดับเลขคณิตของจำนวนเต็มบวกซึ่ง

$$\sum_{i=1}^{7} a_{b_i} = 7 \left(\sum_{i=1}^{7} b_i \right) = 2009$$

แล้ว $\sum_{i=1}^{81} a_i$ มีค่าเท่าใด $\left(\textit{\textbf{Hint}} : \sum_{i=1}^{81} a_i = 81 \cdot a_{41} \right)$